Functional Annotation of *moaC* (Q88NC0) in *Pseudomonas putida* KT2440

Gene: *moaC* (ordered locus PP_1292) **Protein:** Cyclic pyranopterin monophosphate synthase (EC 4.6.1.17) **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) **UniProt:** Q88NC0 · Family: MoaC (HAMAP MF_01224) · Length: 156 aa

1. Summary (Answer to the Research Question)

moaC encodes **cyclic pyranopterin monophosphate (cPMP) synthase**, a cytoplasmic lyase that catalyzes the second half of the **first, committed step of molybdenum cofactor (Moco) biosynthesis**. Working immediately downstream of the radical-SAM enzyme MoaA, MoaC converts the cyclic nucleotide **(8S)-3′,8-cyclo-7,8-dihydroguanosine 5′-triphosphate (3′,8-cH_2GTP)** into **cyclic pyranopterin monophosphate (cPMP, "precursor Z")**, releasing diphosphate (EC 4.6.1.17). It is this reaction that builds the characteristic **pyranopterin ring** and the terminal **cyclic phosphate** that ultimately coordinate molybdenum in every molybdoenzyme. The enzyme functions as a **homohexamer (trimer of dimers)** in the **cytoplasm**, with composite active sites at subunit interfaces.

The gene identity is unambiguous: the UniProt catalytic annotation, the MoaC/HAMAP family assignment, the conserved active-site motif, and the InterPro domain complement all agree, and the P. putida protein is a canonical, full-length MoaC ortholog of the extensively characterized *E. coli* enzyme.

2. Gene/Protein Identity Verification

Attribute	Finding
Protein name	Cyclic pyranopterin monophosphate synthase (UniProt recommended name)
EC number	4.6.1.17 (a carbon–oxygen lyase acting on phosphates)
Family / rule	MoaC family; HAMAP MF_01224
InterPro domains	MoaC (IPR023045), MoaC_bact/euk (IPR047594), MoaC_sf (IPR036522), MoaA/MoaC (IPR050105), Molybdenum-cofactor-biosynthesis-C domain (IPR002820)
Length	156 aa (typical for bacterial MoaC, ~17 kDa)
Active-site motif	Conserved cyclohydrolase-type "...L**CHPL**M..." segment present in the sequence
Catalytic reaction (UniProt)	3′,8-cH_2GTP = cyclic pyranopterin phosphate + diphosphate
Quaternary structure (UniProt)	Homohexamer, trimer of dimers

Conclusion: Q88NC0 is a bona fide MoaC. All literature cited below concerns the same MoaC / cPMP-synthase protein family (bacterial orthologs and the human homolog MOCS1B), so there is no gene-symbol ambiguity for this target.

Quantitative homology evidence (this work): A pairwise alignment of Q88NC0 against the experimentally and structurally characterized *E. coli* MoaC (P0A738) shows **72.7% amino-acid identity** (112/154 aligned positions), and the catalytically essential *E. coli* **Asp128 aligns with an aspartate in the P. putida sequence**. KEGG independently assigns PP_1292 to ortholog **K03637** (cyclic pyranopterin monophosphate synthase, EC 4.6.1.17), Pfam **MoaC**, and module **M00880** ("Molybdenum cofactor biosynthesis, GTP => molybdenum cofactor"). This high identity plus conservation of the catalytic residue provides strong homology-based support for transferring the *E. coli* function to this specific *P. putida* protein.

Genomic context (this work): PP_1292 (*moaC*, 1,478,937–1,479,407) lies in a contiguous, same-strand **moaC–moaD–moaE** cluster: PP_1293 = *moaD* (molybdopterin synthase small subunit) and PP_1294 = *moaE* (molybdopterin synthase large subunit). *moaC* and *moaD* overlap by ~4 bp and *moaD*/*moaE* by 3 bp, indicating **translational coupling / operonic organization**. Thus the gene for pathway step 1 (MoaC → cPMP) is co-located and co-transcribed with the genes for step 2 (MoaD/MoaE, cPMP → molybdopterin) — genomic evidence that MoaC's product is directly handed to the downstream enzymes of Moco biosynthesis.

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3. Primary Molecular Function

3.1 The reaction catalyzed

The first step of Moco biosynthesis is a remarkable transformation of GTP into cPMP that is carried out by **two** enzymes acting in sequence:

1. **MoaA** (a radical *S*-adenosyl-methionine enzyme with two [4Fe-4S] clusters) abstracts a hydrogen atom and forms a new C–C bond, converting GTP into the cryptic cyclic intermediate **3′,8-cH$_2$GTP** [Hover 2013, P 23627491; Hänzelmann & Schindelin 2004/2006].
2. **MoaC** then catalyzes the **complex intramolecular rearrangement** of 3′,8-cH$_2$GTP that forms the pyranopterin ring and the cyclic phosphate, yielding **cPMP** and releasing diphosphate [Hover 2013, P 23627491; Hover 2015, P 26575208].

Historically MoaC was thought to play only an accessory role (pyrophosphate release / cyclic phosphate formation). The isolation of the MoaA product 3′,8-cH$_2$GTP and its efficient conversion to cPMP by MoaC (and by the human homolog MOCS1B) **redefined MoaC as the enzyme that performs the majority of the pyranopterin-forming chemistry** — "in contrast to previous proposals, MoaC plays a major role in the complex rearrangement to generate the pyranopterin ring" [Hover 2013, P 23627491].

3.2 Substrate specificity

MoaC is highly specific for the cyclic nucleotide substrate: 3′,8-cH_2GTP is converted to cPMP with **Km < 0.060 μM** for bacterial MoaC (0.79 ± 0.24 μM for human MOCS1B) [Hover 2013, 23627491].

Mechanistic probing with an uncleavable substrate analogue (3′,8-cH_2GMP[CH_2]PP) showed that the early catalytic steps proceed **before** cyclic-phosphate formation, and that partially active MoaC variants generate a defined reaction intermediate — providing direct evidence on the ordering of bond-forming events in the pyranopterin rearrangement [Hover 2015, 26575208].

3.3 Catalytic residues

The *E. coli* MoaC crystal structure identified a conserved active-site pocket at the dimer interface; **Asp128** is catalytically essential (Asp128→Ala nearly abolishes activity), and a substitution in the human ortholog (Thr182→Pro) that causes Moco deficiency maps to the same region [Wuebbens 2000, 10903949].

GTP-bound MoaC structures (e.g., from *Thermus thermophilus*) further map the substrate-binding residues [Kanaujia 2010, 20606263].

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4. Structure and Localization

- **Fold:** Each monomer adopts a **ferredoxin-like fold** — a four-stranded antiparallel β-sheet packed against two long α-helices [Wuebbens 2000, 10903949].
- **Oligomer:** The enzyme is a **tightly packed homohexamer with 32 symmetry (trimer of dimers)**; the active sites lie **at the interface between two monomers**, so oligomerization is required to assemble competent catalytic pockets [Wuebbens 2000,

P 10903949 ;

UniProt Q88NC0].

- **Localization:** MoaC catalyzes an intracellular reaction and is part of the **soluble cytoplasmic** Moco-biosynthesis machinery; the pathway operates in the cytoplasm of the bacterial cell [Mendel & Leimkühler 2015, P 24980677]. The mature Moco is subsequently inserted into apo-molybdoenzymes, some of which are exported to the periplasm, but the MoaC reaction itself is cytoplasmic.
- **Conservation:** MoaC-type proteins are present across archaea, bacteria and eukaryotes; structural studies of orthologs (e.g., *Mycobacterium tuberculosis* MoaC2) confirm the conserved fold and interaction with MoaA [Srivastava 2013, P 23526978].

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5. Pathway Context and Biological Role

MoaC catalyzes **step 1 of the four-step bacterial Moco pathway** [Mendel & Leimkühler 2015, P 24980677 ; Schwarz & Mendel 2006, P 16669776]:

1. **GTP → cyclic pyranopterin monophosphate (cPMP)** — MoaA + **MoaC** ← *this gene*
2. cPMP → molybdopterin (MPT) — MoaD/MoaE (MPT synthase), sulfuration by MoeB
3. Mo insertion into MPT → Moco — MogA/MoeA
4. (bacteria/archaea) nucleotide attachment → bis-MGD / MCD dinucleotide variants of Moco

Moco is the essential catalytic cofactor of >**50 molybdoenzymes**, in which a **pyranopterin coordinates the molybdenum atom at the active site** [Magalon & Mendel 2015, P 26435257]. These enzymes carry out key oxygen-atom-transfer and redox reactions in global carbon, nitrogen and sulfur metabolism — e.g., **nitrate reductase, DMSO/TMAO reductase, sulfite**

oxidase, xanthine dehydrogenase/oxidase, and formate dehydrogenase. In a metabolically versatile soil bacterium such as *P. putida* KT2440, Moco-dependent enzymes support **alternative/anaerobic respiration and nitrogen and sulfur metabolism**; MoaC is therefore the gatekeeping biosynthetic step that activates all of this molybdenum-dependent chemistry. Because cPMP synthesis is the entry point of the pathway, loss of MoaC function would abolish Moco production and inactivate the entire downstream molybdoenzyme repertoire. In humans, defects in the orthologous first step cause fatal **Molybdenum Cofactor Deficiency**, underscoring the step's essentiality [Wuebbens 2000, P 10903949].

Pathway completeness in P. putida KT2440 (this work): KEGG ortholog mapping confirms the organism encodes the entire Moco biosynthetic machinery — *moaA* (three paralogs: PP_1969, PP_2482, PP_4597), *moaC* (single copy, PP_1292), *moaD* (PP_1293), *moaE* (PP_1294), *moeA* (PP_2123, Mo insertase) and *mogA* (PP_3457) — i.e., every step needed to convert GTP into mature Moco (KEGG module M00880). Notably, **MoaC is present as a single, non-redundant copy**, so PP_1292 is the sole cyclic-pyranopterin-monophosphate synthase in the cell and an obligatory node of the pathway; the three MoaA paralogs contrast with this single MoaC. The presence of all downstream enzymes confirms that MoaC's product (cPMP) is channeled through a complete, functional pathway to activated molybdenum cofactor.

6. Supported and Refuted Hypotheses

Hypothesis	Status	Basis
Q88NC0 is a canonical MoaC / cPMP synthase	**Supported**	UniProt catalytic annotation, HAMAP MF_01224, InterPro domains, conserved CHPL motif, 156-aa length
Substrate is the cyclic nucleotide 3′,8-cH_2GTP (product of MoaA), not free GTP	**Supported**	Hover 2013 (P 23627491)
MoaC performs the pyranopterin ring-forming rearrangement (not just PPi release)	**Supported** (revised historical view)	Hover 2013/2015
MoaC functions as an oligomer with interfacial active sites	**Supported**	Wuebbens 2000 (P 10903949); UniProt homohexamer
The reaction occurs in the cytoplasm as step 1 of Moco biosynthesis	**Supported**	Mendel & Leimkühler 2015; Schwarz & Mendel 2006
MoaC has a molybdenum-independent "moonlighting" primary role	**Not supported**	No evidence; all data point to Moco biosynthesis as the sole primary function
PP_1292 is functionally equivalent to E. coli MoaC	**Supported**	72.7% identity, conserved catalytic Asp, KEGG K03637/M00880 (this work)
PP_1292 is organized in a moa operon in P. putida	**Supported**	Contiguous, overlapping moaC–moaD–moaE cluster (this work)

7. Limitations and Future Directions

- **Organism-specific data:** Direct biochemical/genetic characterization of MoaC has been done in *E. coli*, *T. thermophilus*, *M. tuberculosis* and human MOCS1B, **not specifically in *P. putida* KT2440**. The functional assignment for PP_1292 rests on very high sequence/domain conservation (72.7% identity to *E. coli* MoaC, conserved catalytic Asp), operonic co-location with *moaD/moaE*, and the essentiality of the pathway; a *P. putida* knockout/ complementation study would still provide direct experimental confirmation.

- **Mechanistic details:** The precise atom-by-atom mechanism of the pyranopterin rearrangement is still being resolved; recent radical-SAM (MoaA) work continues to refine the picture [Yokoyama 2022, P 35480226].
- **Regulation/operon context:** The transcriptional regulation and *moa* operon organization of PP_1292 in *P. putida* (and any physiological conditions modulating Moco demand, e.g., nitrate/ anaerobiosis) were not analyzed here and are a natural follow-up.

8. Key References

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